Synthetic Data Generation for Alleviating Class Imbalance with Reinforcement Learning

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***Abstract*—Wearable sensors are increasingly used in daily life for applications such as fitness monitoring, rehabilitation, and workplace safety. Human activity recognition (HAR) models play a central role in these systems by classifying movement patterns. Nevertheless, class imbalance reduces model reliability, particularly in health-related contexts. This paper proposes an adaptive reinforcement learning framework for synthetic data generation that operates as a feedback-driven process. In each training episode, the agent generates minority-class samples, a reference classifier evaluates their utility, and the policy is updated based on both the improvement in minority-class performance and the proximity of samples to decision boundaries. Experimental results show that the proposed method improves minority-class F1 scores over random undersampling, random oversampling, and generative adversarial network baselines, particularly in low-separability scenarios where class overlap is significant. These results demonstrate that feedback-based adaptive synthesis effectively addresses imbalance by targeting underrepresented regions in the feature space, leading to fairer and more reliable activity recognition.**

***Index Terms*—class imbalance, reinforcement learning, human activity recognition, synthetic data, cyber-physical systems**

I. Introduction

Wearable sensors are becoming integral to everyday life, supporting applications ranging from fitness tracking to rehabilitation and workplace safety [1]. Human activity recognition (HAR) models increasingly drive decisions in these settings, classifying a wide range of everyday movements [2]. Within cyber-physical systems (CPS), HAR interprets sensor streams to guide downstream control and decision-support processes in environments such as assisted living and rehabilitation. However, these models are only as reliable as their training data and are susceptible to biases within the training dataset [3]. Common activities such as walking and running are well represented, whereas movements such as jumping rope or climbing stairs occur far less frequently [4]. This imbalance leads to misclassification of minority activities, raising concerns about system reliability in real-world CPS deployments [3].

A prominent way to address data scarcity in HAR is through synthetic data generation, which enables models to learn from additional examples of minority activities without the labor-intensive cost of new data collection [5]. Techniques such as generative adversarial networks (GANs) and variational autoencoders are widely used to augment training datasets with realistic samples [5]. By expanding the diversity of available data, synthetic generation allows models to encounter a broader range of activity patterns during training, improving generalization while reducing dependence on extensive real-world data collection [6].

Generative models have advanced synthetic data generation by incorporating constraints aimed at mitigating imbalance [7], [8]. However, most such methods treat concepts of fairness and imbalance as a static constraint and do not continuously adapt based on how a downstream classifier performs [9]. A few more recent extensions attempt to include downstream task metrics into the generator's objective [10]. Yet these methods still rely on pre-defined objectives instead of learning continuously from feedback.

The proposal argues that data generation should be treated as an adaptive learning process that continuously refines how it produces synthetic samples based on classifier feedback. Reinforcement learning (RL) provides a natural framework for this: by rewarding policies that generate data improving minority-class performance, an RL-based generator can dynamically balance fair representation during training.

The contributions of this paper are as follows:

1) An RL framework, ADAPT-RL, for synthetic data generation that encodes class-specific performance objectives directly into the reward design.
2) Demonstration of how adaptive feedback during training enables the generator to dynamically adjust its output distribution, strengthening representation of underrepresented activities while maintaining realism across the dataset.
3) An empirical evaluation on benchmark HAR data, showing improvements in minority-class performance without sacrificing overall model accuracy.

The remainder of this paper is organized as follows. Section II reviews prior approaches in generative modeling and RL for data augmentation. Section III presents our proposed framework, with Section IV reporting experimental findings. Finally, Section V summarizes contributions and outlines future directions.

II. Related Work

Research on mitigating data scarcity spans multiple areas of machine learning. Key contributions come from generative modeling methods, which synthesize data to balance class distributions, and from RL approaches, which incorporate feedback-driven learning to improve overall performance. This

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section reviews representative works in each area, outlining how they contribute to data augmentation and bias mitigation in imbalanced datasets.

A. Generative Models

Generative Adversarial Networks (GANs): GAN-based methods are widely used to address data scarcity and class imbalance by generating additional samples for underrepresented classes. CTGAN [11] adapted GANs for tabular data, improving the modeling of mixed data types and skewed categorical distributions. DECAF [8] introduced causal constraints to preserve realistic feature relationships and reduce false correlations that contribute to biased representations. Subsequent variants, such as those proposed by Abroshan et al. [9] and FACGAN [12], explicitly regularized the generator to improve minority coverage or reduce class dependency in the latent space.

While GAN methods enhance data diversity, their fixed objectives remain detached from integrated downstream performance, producing plausible but potentially suboptimal samples.

B. Reinforcement Learning Approaches

RL has recently been applied to mitigate representation bias in sequential decision-making. Wen et al. proposed fairness-aware MDPs that constrain policy optimization to prevent over-representation of majority groups and promote balanced outcomes across subpopulations [13]. Claure et al. applied RL to resource allocation, using Q-learning and policy gradient methods to ensure that minority groups are not systematically disadvantaged in scheduling and distribution tasks [14]. RLGen introduced RL for synthetic data generation, rewarding agents for producing feature vectors that improve downstream performance, irrespective of class imbalances [15]. Based on a search of recent work in HAR, synthetic data generation, and fairness-aware RL, no study was found that applies reinforcement learning to generate synthetic data to address class imbalance.

III. Methodology

This section outlines ADAPT-RL as an RL framework for adaptive synthetic data generation. The approach enables an agent to iteratively produce synthetic minority samples, evaluate their effect on downstream classification, and adjust future generation based on feedback. In each episode, a classifier trained on real data serves as a reference for evaluating how newly generated samples influence minority-class performance on a validation set. The feedback is then used to update the agent's policy, forming a closed learning loop that progressively aligns synthetic data generation with representation objectives.

The overall process is illustrated in Fig. 1, which outlines the interaction between the agent, classifiers, and reward computation throughout training. After training concludes, the best-performing synthetic dataset is selected and combined with held-out test data to evaluate final model performance.

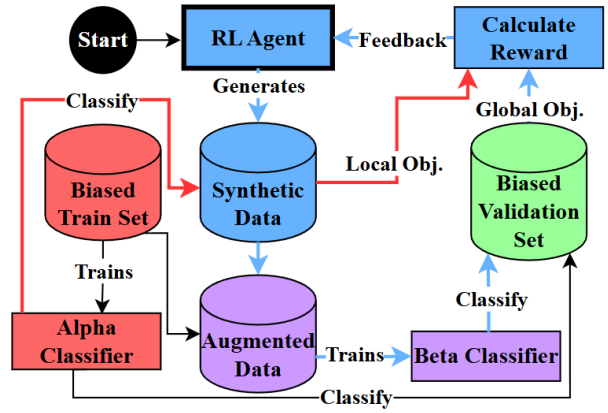

Fig. 1: Training pipeline for the reinforcement learning framework.

A. Problem Formulation

The objective is to improve minority-class recognition in an imbalanced classification setting by learning synthetic feature vectors that enhance downstream classifier performance. Let $\mathcal{D}_{\text{real}} = \{(\mathbf{x}_i, y_i)\}$ denote the real training data, where $y_i \in \{0, 1\}$ correspond to majority and minority labels. The task is to generate a set of synthetic samples $\Phi = \{\tilde{\mathbf{x}}_j, \tilde{y}_j{=}1\}$ such that classifier performance on a validation set θ_{val} is maximized according to:

$$\max_{\Phi} \mathrm{F1}_{\text{minority}}(\mathcal{D}_{\text{real}} \cup \Phi \mid \theta_{\text{val}}). \tag{1}$$

This formulation defines the learning goal: discovering synthetic samples that close the representation gap between minority and majority classes by directly optimizing minority-class performance. The following subsection describes how this objective is implemented using an RL formulation with adaptive reward feedback.

B. Environment Implementation

The environment consists of two classifiers: a fixed baseline classifier α trained on real data, and an episode classifier β retrained each episode using the real data combined with the synthetic samples generated by the agent.

1) Agent and Action: The agent corresponds to a generator parameterized by a stochastic policy π_θ, which is optimized using the REINFORCE policy-gradient algorithm. Given a state s_t, it produces a continuous action $a_t \in \mathbb{R}^d$ representing a synthetic feature vector automatically labeled as the minority class. Across T steps per episode, the agent constructs a synthetic batch $\Phi = a_1, a_2, \ldots, a_T$ that is merged with the real dataset for evaluation.

2) Objective Function: To align the data generation with the objective in Eq. (1), two objective functions, combined into a trajectory reward, are defined in Eq. (2):

$$R_k = \lambda_k r_k^{\text{global}} + (1 - \lambda_k) r_t^{\text{local}}, \tag{2}$$

where r_k^{global} is a global objective that measures overall performance improvement over an episode trajectory k. r_t^{local}

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is the local objective that promotes exploration in regions most informative for classification over timestep t. And $\lambda_k \in [0, 1]$ is the weight over the rewards that linearly decays across training episodes to gradually shift the emphasis from local exploration to global performance improvement.

Global reward is calculated each episode and captures whether the generated data improves minority-class classification compared to the baseline classifier by:

$$r_k^{\text{global}} = \text{F1}_{\text{minority}}(\beta \mid \theta_{\text{val}}) - \text{F1}_{\text{minority}}(\alpha \mid \theta_{\text{val}}) \quad (3)$$

where α is the baseline classifier trained with the imbalanced data and β is the second classifier trained with the augmented data.

The local reward is obtained at each timestep and offers per-action feedback that incentivizes the agent to concentrate on regions of uncertainty within the classifier's decision space. The intuition is that samples near the classifier's decision boundary, where its predictions are close to 0.5, are the most informative for improving future classification performance. Consequently r^{local} is formalized as:

$$r_t^{\text{local}} = \exp\left(-\frac{1}{2}\left(\frac{|p_\alpha(y{=}1 \mid \mathbf{x}_t) - 0.5|}{\tau}\right)^2\right), \quad (4)$$

where $p_\alpha(y{=}1 \mid \mathbf{x}_t)$ is the predicted probability of the minority class from the baseline classifier α. τ controls the region of uncertainty and $\tau > 0$. Samples with probabilities near 0.5 receive the highest reward, while confidently classified samples near 0 or 1 yield lower rewards, as they contribute less new information.

IV. Results

A. Experimental Setup

Experiments were conducted on a Linux server (AMD 3970X CPU, 128 GB RAM, RTX 3090 GPU) using Python 3.10 and PyTorch 2.6.0 with CUDA 12.4. All results represent the mean of three runs with different random seeds. To assess the effectiveness of the proposed RL-based generator in mitigating class imbalance, experiments were conducted against three baseline methods: random undersampling (RUS), random oversampling (ROS), and CTGAN [11]. RUS balances classes by reducing majority samples, while ROS duplicates minority samples to equalize class frequencies. CTGAN, a generative adversarial network for tabular data, synthesizes new samples by modeling complex feature dependencies. All baselines employed identical classifier architectures, preprocessing, and data splits to ensure a fair comparison with the proposed approach.

B. Dataset Analysis and Scenario Configuration

The PAMAP2 human activity dataset [16] was employed, consisting of multimodal sensor data from wearable inertial and heart-rate devices. Sensor streams were segmented into five-second windows with a 2.5-second overlap, normalized to zero mean and unit variance, and reduced using principal component analysis to lower input dimensionality for the RL agent. The task was formulated as a binary classification problem with an induced imbalance by downsampling the minority class. Two scenarios were designed to evaluate the framework under different separability conditions: Scenario 1, a high-separability case based on distinct motion activities; and Scenario 2, a low-separability case where similar motion patterns create overlapping feature distributions and increased classification difficulty. Figure 2 shows the scenario analysis with the different activities selected.

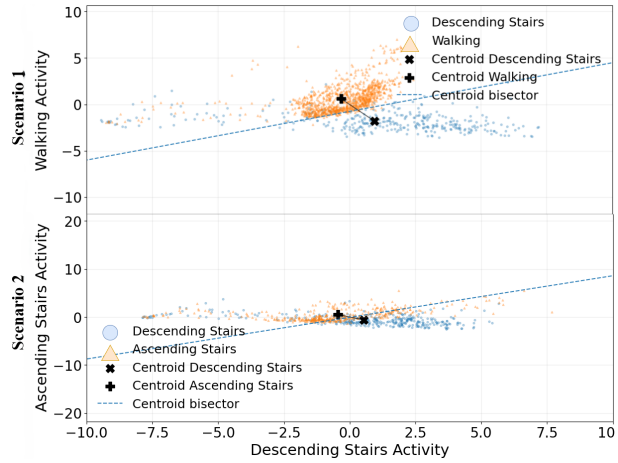

Fig. 2: Separability analysis for scenarios configuration.

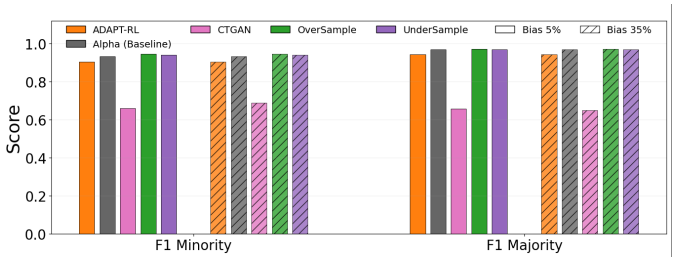

Fig. 3: Experiment 1: performance in high-separability scenario.

C. Experiment 1: High-Separability Case

Experiment 1 evaluated ADAPT-RL on the high-separability scenario, where the minority activity *descending stairs* was classified against the majority activity *walking*. These activities exhibit distinct motion characteristics and sensor signatures, leading to clear feature boundaries that facilitate class discrimination. The evaluation was conducted under two imbalance levels (5% and 35% minority prevalence), reporting minority, majority, and weighted F1 scores averaged over three random seeds. Results are summarized in Figure 3.

D. Experiment 2: Low-Separability Case

Experiment 2 addressed a more challenging low-separability scenario, pairing the minority activity descending stairs with the majority activity ascending stairs. These two classes share similar movement dynamics and sensor readings, resulting in substantial overlap in the feature space and making discrimination difficult for traditional balancing methods. Experiments were performed using the same imbalance levels and evaluation

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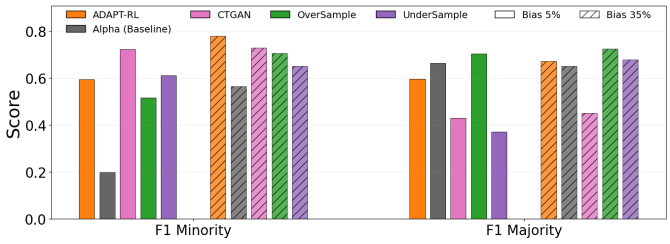

Fig. 4: Experiment 2: performance in low-separability scenario.

metrics as in the previous case. Figure 4 shows the results for experiment 2.

E. Discussion

Across both evaluation scenarios, the results demonstrate that class separability is a critical factor influencing the performance of balancing strategies. In the high-separability setting (*descending stairs* vs. *walking*), all methods achieved comparable results, suggesting that conventional resampling techniques are sufficient when class boundaries are well defined. Conversely, in the low-separability setting (*descending stairs* vs. *ascending stairs*), ADAPT-RL outperformed random undersampling, oversampling, and CTGAN by attaining higher minority F1 scores or lower tradeoffs with majority performance.

V. Conclusion and Future Work

This work explored ADAPT-RL for imbalanced HAR using reinforcement learning. The proposed method formulates data generation as a feedback-driven process, where an agent iteratively proposes minority-class samples, a classifier evaluates their contribution to improving performance, and reward signals guide subsequent generation. Experimental results demonstrated that this approach achieved competitive results in high-separability settings and improved minority-class recognition in low-separability scenarios compared to random undersampling, random oversampling, and CTGAN. These findings suggest that coupling data generation with downstream feedback enables the model to more effectively target underrepresented regions of the decision space than static resampling or unconditional synthesis, particularly when overlapping features make minority recovery challenging.

The present study, however, has certain limitations. It focused on binary activity discrimination using tabularized data segments, which simplifies the temporal dependencies inherent in raw sensor streams. Training the model introduces additional computational cost due to repeated classifier retraining and sensitivity to reward scaling and scheduling. Moreover, while dimensionality reduction was applied to stabilize training and reduce complexity, its effect on generalization to higher-dimensional inputs remains unexplored. Future work will extend this framework to multi-class activity recognition and longer temporal contexts, integrating sequence models capable of operating directly on time-series windows. Additional research will also explore hybrid objectives that balance task performance with distributional realism, adaptive reward schedules, uncertainty-aware rewards, and curriculum strategies for varying imbalance levels. Broader validation across multiple HAR datasets and deployment in CPS will further assess the framework's generalizability and practical utility.

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